Literature search: queries

Methodology of Research, assignment 2 · Eduard Kecskemeti · topic: estimating “apparent weeks out” from physique photos (Classic Physique) with pairwise ranking · searches run 17–19 Oct 2026

1. Search set-up

Item	Setting
Research question	How has progress toward contest leanness, or body composition in general, been estimated from ordinary photographs?
Sub-questions	Q1 body composition / shape from 2D images · Q2 body-composition tracking during contest preparation · Q3 learning an order or progression from images without absolute labels
Term refinement	Google Scholar Labs: each sub-question asked in natural language; results used only to collect vocabulary and seed papers
Database	OpenAlex (api.openalex.org), field **title_and_abstract.search**. Web of Science: login failed. Scopus: institutional account pending, so only Scopus Preview. Scholar has no type/language filters or export.
Filters (all queries)	`publication_year:2016-2025, language:en, type:article|review|conference-paper|book-chapter`
Criteria	Articles (incl. reviews), conference papers, book chapters; English; 2016–2025, current year (2026) excluded

2. Final queries (used for the dataset)

Block	Q	Query	Hits
b1v3	Q1	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry") AND ("deep learning" OR "machine learning" OR "neural network" OR convolutional OR "artificial intelligence" OR "computer vision" OR automated OR "2D image" OR "two-dimensional")`	804
b2	Q2	`(bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") AND ("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")`	294
b3v4	Q3	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "siamese network" OR "siamese neural network" OR "temporal order" OR "image ordering") AND (longitudinal OR progression OR "age estimation" OR "temporal change" OR "disease severity") AND (image OR images OR photograph OR imaging)`	138
b4	Q1–Q2	`(bodybuilding OR bodybuilder OR "physique athlete" OR "physique competitor") AND ("computer vision" OR "deep learning" OR "machine learning" OR "artificial intelligence" OR "pose estimation")`	18
		Union of the four queries (unique records)	**1,251**

Block b1 (the original Q1 query, 436 hits, see section 3) is a subset of b1v3 and adds no records; it is kept as a label on the records it matches.

3. Scrapped queries and why they were changed

Version	Query	Hits	Problem → change
Run 1 (all blocks)	Queries b1, b2 and b3 (original), with types `article|review|book-chapter` only	881 union	Conference papers are a separate OpenAlex type and were excluded; most ML work for Q1/Q3 is published at conferences → added `conference-paper`.

Version	Query	Hits	Problem → change
b1	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry") AND ("deep learning" OR "machine learning" OR "neural network" OR convolutional)`	436 (302 in run 1)	Missed seed papers that describe the method without ML words (Majmudar 2022, Ferreira 2025) → third concept widened (b1v3).
b1 without 3rd concept	`("body fat" OR "body composition" OR adiposity OR "body shape" OR physique OR "body mass index" OR anthropometry OR anthropometric) AND (photograph OR photo OR "2D image" OR "single image" OR "RGB image" OR smartphone OR "computer vision" OR "image-based" OR silhouette OR "digital anthropometry")`	3,991	Recall fine but far over the 1,500 limit and mostly clinical / apparel noise → rejected.
b3 (original)	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "relative attribute" OR "temporal order" OR "image ordering" OR siamese) AND (longitudinal OR progression OR "disease stage" OR "temporal change" OR "age estimation") AND (image OR images OR photograph OR "deep learning")`	292	Bare *siamese* matched Thai/"Siam" studies; *progression* without an image term matched oncology → rewritten.
b3v2*	b3 with the third concept replaced by a required neural-network term ("deep learning" / "neural network" family)	46	Too narrow: lost known Q3 papers → rejected.
b3v3*	`("ordinal regression" OR "learning to rank" OR "pairwise ranking" OR "siamese network" OR "siamese neural network" OR "temporal order" OR "image ordering") AND (longitudinal OR progression OR "age estimation" OR "temporal change" OR "disease severity")`	703	Specific network phrases fixed the Siam noise, but with no image term most hits were non-imaging → image concept added (b3v4).

** The exact wording of b3v2 was not preserved; it is described here from the search log. b3v3 is b3v4 without its image concept, as recorded in the log.*

4. From hits to datasets

Step	Records	Notes
Union of final queries	1,251	
Cleaning	− 84	Spam ads 11 · retractions 5 · front matter / abstract books 21 · no DOI and no source 2 · repository copies and preprints 31 · non-English titles 5 · duplicate titles 9
Snowballing	+ 4	Scholar Labs seeds not matched by any query: Liu 2018 (CVPR), Minetto 2023 (Sci. Rep.), Kasper 2021 (Nutrients), Toulany 2023 (Nat. Methods)
Dataset 1 (broad)	**1,171**	879 articles, 17 reviews, 242 conference papers, 33 book chapters; under the 1,500 limit
Dataset 2 (relevant)	**50**	Screened from dataset 1; 46 from the search, 4 from snowballing; themes: photo-based body composition 15, shape/weight from images 9, contest prep 13, ranking/ordering 11, AI in bodybuilding 2

5. Planned re-run in Scopus / Web of Science

When access is approved the four final queries will be run with the same concepts and filters, merged with the OpenAlex set by DOI, and dataset 2 re-screened. Scopus form (b2 shown; the others follow the same pattern):

```
TITLE-ABS-KEY((bodybuilder OR bodybuilding OR "physique athlete" OR "physique competitor" OR "physique competition" OR "contest preparation" OR "competition preparation") AND ("body composition" OR "body fat" OR "fat loss" OR "weight loss" OR "energy restriction" OR "lean mass" OR "fat mass")) AND PUBYEAR > 2015 AND PUBYEAR < 2026 AND LANGUAGE(english) AND DOCTYPE(ar OR re OR cp OR ch)
```

Web of Science: the same string in the **TS=** field, years 2016–2025, document types Article, Review, Proceedings Paper, Book Chapter, language English.